Programming in c [ppc18]

Unit-1

Introduction to computers

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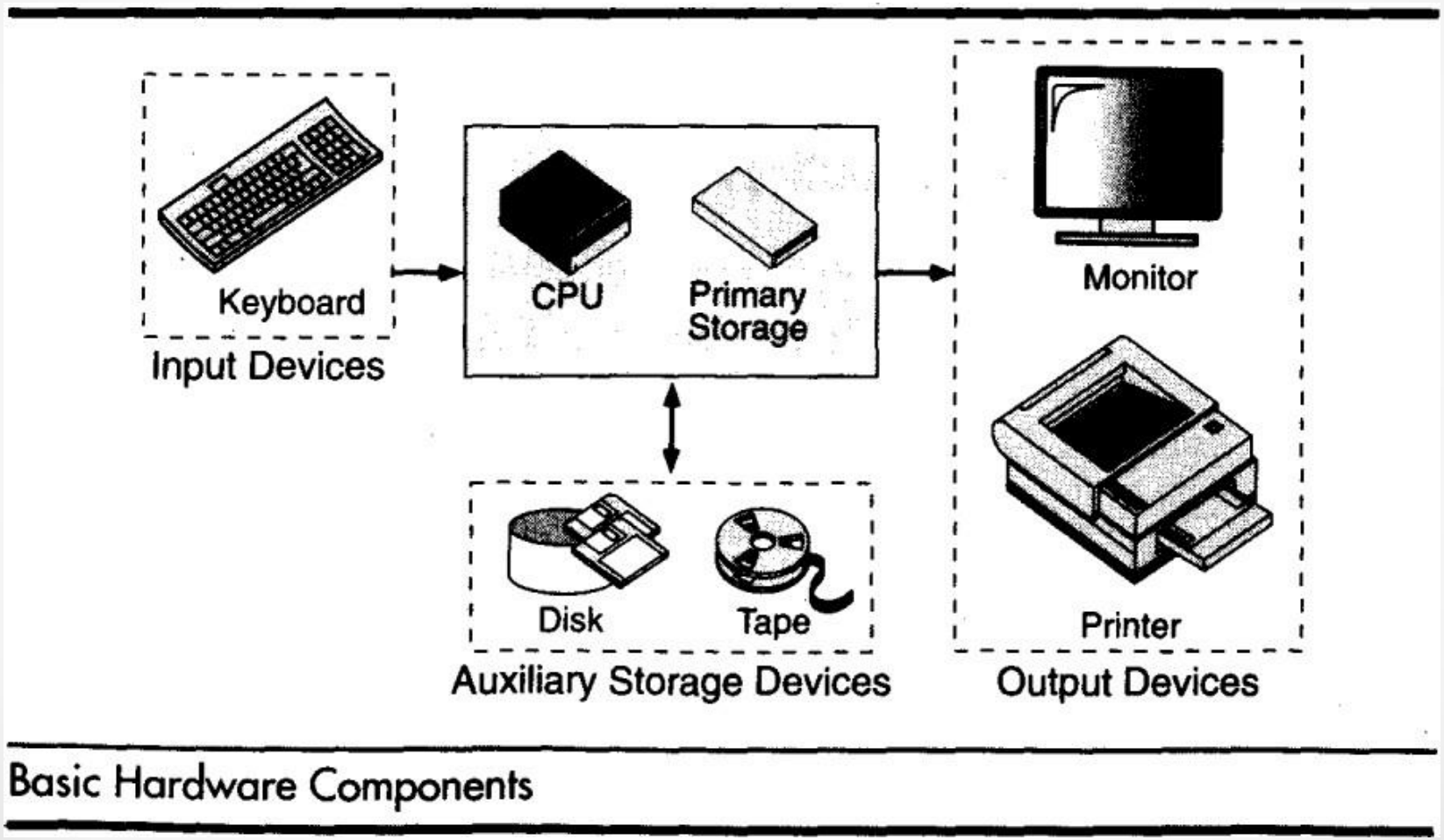
Computer Systems:

- Computer system is a programmable Electronic device that can accept Input,store data & retrieve ,process & output Information.
- Computer is a System made of 2 major components
  1. Hardware
  2. Software
- The Computer Hardware is physical Equipment.
- The Software is a collection of programs (instructions) that allows hardware to do its job

Computer Hardware

- The hardware component of the computer system consists of five parts: input devices, central processing unit (CPU) ,primary storage, output devices, and auxiliary storage devices.

Computer Hardware



Basic Hardware Components

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- The **input device** is usually a keyboard where programs and data are entered into the computers. Examples of other input devices include a mouse, a pen or stylus, a touch screen, or an audio input unit.
- The **central processing unit (CPU**) is responsible for executing instructions such as arithmetic calculations, comparisons among data, and movement of data inside the system. Today's computers may have one ,two, or more CPUs .
- **Primary storage** ,also known as main memory, is a place where the programs and data are stored temporarily during processing. The data in primary storage are erased when we turn off a personal computer or when we log off from a time-sharing system.
- The **output device** is usually a monitor or a printer to show output. If the output is shown on the monitor, we say we have a **soft copy**. If it is printed on the printer, we say we have a hard copy.
- **Auxiliary storage**, also known as **secondary storage**, is used for both input and output. It is the place where the programs and data are stored permanently. When we turn off the computer, or programs and data remain in the secondary storage, ready for the next time we need them.

Computer Software

- Computer software is divided in to two broad categories: system software and application software .System software manages the computer resources .It provides the interface between the hardware and the users. Application software, on the other hand is directly responsible for helping users solve their problems

Types Of Computer Software

- System Software
- Application Software

System Software:

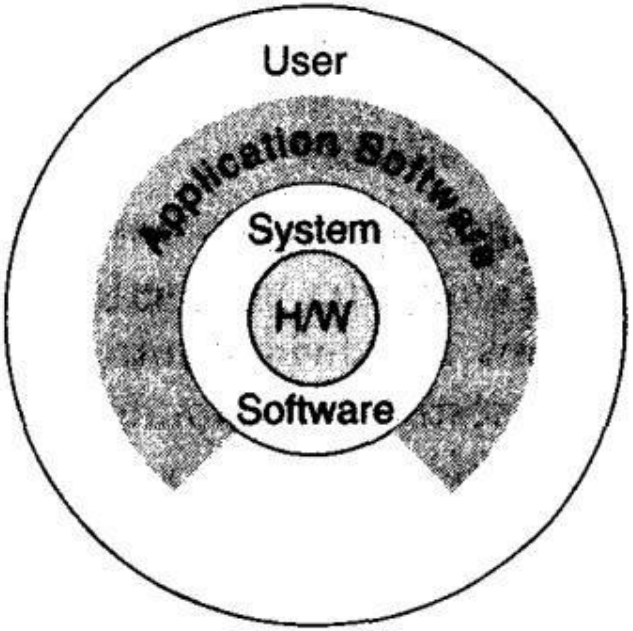
- **System software** consists of programs that manage the hardware resources of a computer and perform required information processing tasks. These programs are divided into three classes: the operating system, system support, and system development.
- The **operating system** provides services such as a user interface, file and database access, and interfaces to communication systems such as Internet protocols. The primary purpose of this software is to keep the system operating in an efficient manner while allowing the users access to the system.
- **System support software** provides system utilities and other operating services. Examples of system utilities are sort programs and disk format programs. Operating services consists of programs that provide performance statistics for the operational staff and security monitors to protect the system and data.
- The last system software category ,**system development software**, includes the language translators that convert programs into machine language for execution ,debugging tools to ensure that the programs are error free and computer –assisted software engineering(CASE) systems.

Application software

- **Application software** is broken in to two classes :general-purpose software and application – specific software. **General purpose software** is purchased from a software developer and can be used for more than one application. Examples of general purpose software include word processors ,database management systems ,and computer aided design systems. They are labeled general purpose because they can solve a variety of user computing problems.
- **Application –specific software** can be used only for its intended purpose.

- A general ledger system used by accountants and a material requirements planning system used by a manufacturing organization are examples of application-specific software. They can be used only for the task for which they were designed they cannot be used for other generalized tasks.
- The relationship between system and application software is shown in fig.In this figure, each circle represents an interface point .The inner core is hard ware. The user is represented by the out layer. To work with the system, the typical user uses some form of application software. The application software in turn interacts with the operating system , which is apart of the system software layer. The system software provides the direct interaction with the hard ware. The opening at the bottom of the figure is the path followed by the user who interacts directly with the operating system when necessary.

Relationship b/n system and Application Software



Relationship Between System and Application Software

Computer Languages:

- To write a program for a computer, we must use a **computer language.** Over the years computer languages have evolved from machine languages to natural languages.
- 1940's Machine level Languages
- 1950's Symbolic Languages
- 1960's High-Level Languages

Machine Languages

- In the earliest days of computers, the only programming languages available were machine languages. Each computer has its own machine language, which is made of streams of 0's and 1's.
- Instructions in machine language must be in streams of 0's and 1's because the internal circuits of a computer are made of switches transistors and other electronic devices that can be in one of two states: off or on. The off state is represented by 0 , the on state is represented by 1.
- The only language understood by computer hardware is machine language.

Symbolic Languages:

- In early 1950's Admiral Grace Hopper, A mathematician and naval officer developed the concept of a special computer program that would convert programs into machine language.
- The early programming languages simply mirror to the machine languages using symbols of mnemonics to represent the various machine language instructions because they used symbols, these languages were known as symbolic languages.
- Computer does not understand symbolic language it must be translated to the machine language. A special program called assembler translates symbolic code into machine language. Because symbolic languages had to be assembled into machine language they soon became known as assembly languages.
- Symbolic language uses symbols or mnemonics to represent the various ,machine language instructions.

High Level Languages:

- Symbolic languages greatly improved programming efficiency; they still required programmers to concentrate on the hardware that they were using. Working with symbolic languages was also very tedious because each machine instruction has to be individually coded. The desire to improve programmer efficiency and to change the focus from the computer to the problem being solved led to the development of high-level language.
- High level languages are portable to many different computers, allowing the programmer to concentrate on the application problem at hand rather than the intricacies of the computer. High-level languages are designed to relieve the programmer from the details of the assembly language. High level languages share one thing with symbolic languages, They must be converted into machine language. The process of converting them is known as compilation.
- The first widely used high-level languages, FORTRAN (Formula Translation)was created by John Backus and an IBM team in 1957;it is still widely used today in scientific and engineering applications. After FORTRAN was COBOL(Common Business-Oriented Language). Admiral Hopper was played a key role in the development of the COBOL Business language.
- **C is a high-level language used for system software and new application code.**

Creating & Running Programs

- It is the job of Programmer to write & test the Program. There are 4 steps in this Process.

1. Writing & Editing the program
2. Compiling the program
3. Linking the program with required library modules
4. Executing the Program

- For C Programs use text Editor to write the program and save program with filename.c.
- For compiling use CC filename.c
- For Executing use ./a.out

Computing Environment

- **Definition:**
- Computing environment is a collection of computers/machines, software and networks that support the processing and exchange of electronic information.
  Different types of computing environments
- There are **four types** of computing environments. They are:
- Personal computing environment
- Time-Sharing environment
- Client -Server environment
- Distributed environment

Personal Computing Environment:

- A personal computer(PC) is a general purpose computer. The size and capabilities of a personal computer makes it useful for end-users.
- In a personal computer, all the hardware components are tied together, so that it can be used as per our need.

Time Sharing Environment

- Employees in large companies often work in a time sharing environment. In this environment, many users are connected to one or more computers.
- These are minicomputers or central mainframes.
- The terminals used are not programmable, but now we see more microcomputers that are used to simulate the environments terminals.
- In the time sharing environment, the output devices and auxiliary storage are shared by all the users.
- In time sharing environment, each computer must be operated by the central computer.
- In this type of environment, computing is operated by central computer. This central computer has many duties which keeps the computer busy.

Time Sharing Environment

Client Server Environment

- Client/Server is a distribute computing model, in which the client requests services from the server.
- Client/Server usually run on separate computers connected by a network.
- A client is a process or an application that sends messages to a server through the network.
- These messages ask the server to perform a specific task, like finding a customer record in a database.
- Severs usually run on powerful computers, workstations or mainframes
- The administration of the installed equipment is more expensive than maintaining a centralized system

Contd..


Server
Central Storage
Shared Printers
Client 1
Client 2
Client 3
Client 4
Client-Server Environment

Distributed Computing Environment

- A distributed computing environment allows smooth integration of computing functions between different internet server's clients and provides the connection to different servers around the world.
- Distributed Computing is used by distributed systems to solve computational problems
- In Distributed Computing, each problem is divided into several tasks, which are solved by one or more computers.

Contd..

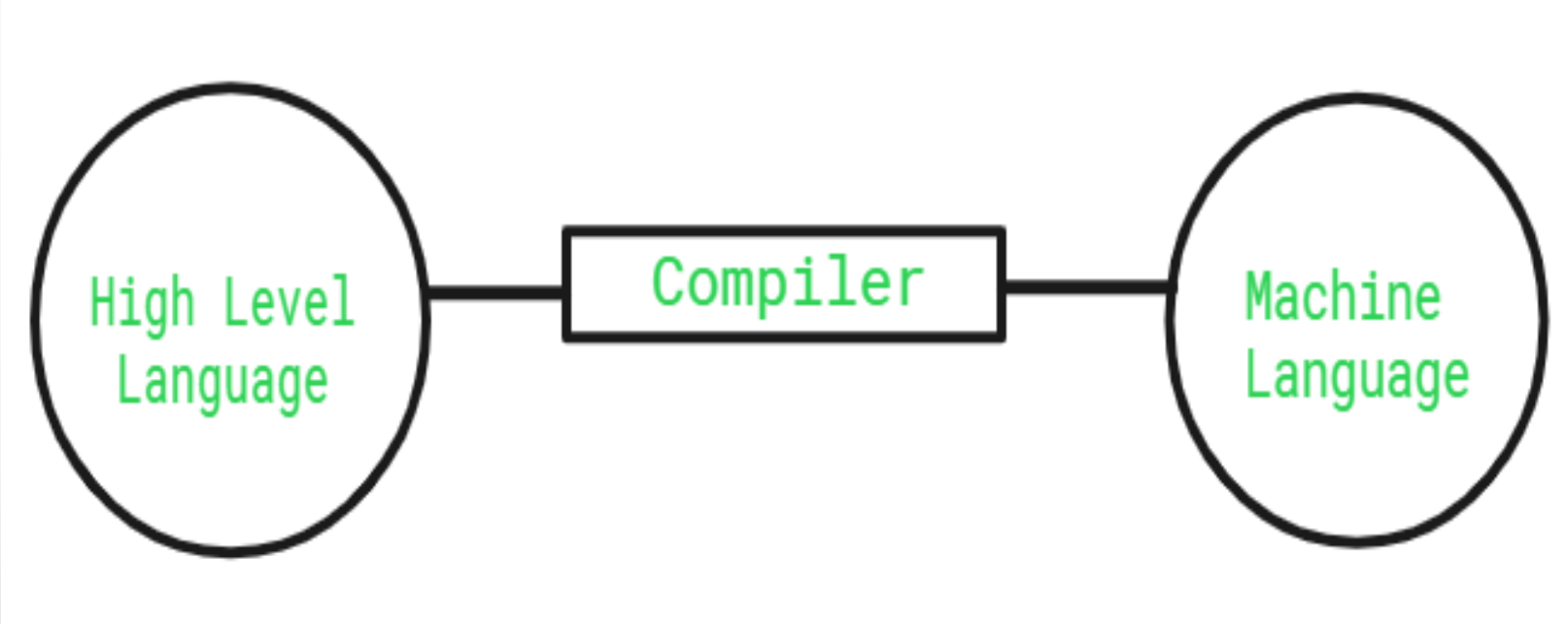

Server
Storage
INTERNET
Storage
Client 1
Client 2
Client 3
Client 4
Clients
(Browsers)
Distributed Computing Environment

Compiler & Interpreter

- The Compiler is a translator which takes input i.e., High-Level Language, and produces an output of low-level language i.e. machine or assembly language. The work of a Compiler is to transform the codes written in the programming language into machine code (format of 0s and 1s) so that computers can understand.
- A compiler is more intelligent than an assembler it checks all kinds of limits, ranges, errors, etc.
- But its program run time is more and occupies a larger part of memory. It has a slow speed because a compiler goes through the entire program and then translates the entire program into machine codes.

Role of a Compiler

- For Converting the code written in a high-level language into machine-level language so that computers can easily understand, we use a compiler. Converts basically convert high-level language to intermediate assembly language by a compiler and then assembled into machine code



Adv & Disadv of Compiler

- **Advantages of Compiler**
- Compiled code runs faster in comparison to Interpreted code.
- Compilers help in improving the security of Applications.
- As Compilers give Debugging tools, which help in fixing errors easily.
- **Disadvantages of Compiler**
- The compiler can catch only syntax errors and some semantic errors.
- Compilation can take more time in the case of bulky code.

Interpreter

- An Interpreter is a program that translates a programming language into a comprehensible language. The interpreter converts high-level language to an intermediate language. It contains pre-compiled code, source code, etc.
- It translates only one statement of the program at a time.
- Interpreters, more often than not are smaller than compilers.
- **Role of an Interpreter**
- The simple role of an interpreter is to translate the material into a target language. An Interpreter works line by line on a code. It also converts high-level language to machine language.

Adv & Disadv of Interpreter

- **Advantages of Interpreter**
- Programs written in an Interpreted language are easier to debug.
- Interpreters allow the management of memory automatically, which reduces memory error risks.
- Interpreted Language is more flexible than a Compiled language.
- **Disadvantages of Interpreter**
- The interpreter can run only the corresponding Interpreted program.
- Interpreted code runs slower in comparison to Compiled code.

Difference Between Compiler and Interpreter

Compiler	Interpreter
The compiler saves the Machine Language in form of Machine Code on disks.	The Interpreter does not save the Machine Language.
Compiled codes run faster than Interpreter.	Interpreted codes run slower than Compiler.
Linking-Loading Model is the basic working model of the Compiler.	The Interpretation Model is the basic working model of the Interpreter.
The compiler generates an output in the form of (.exe).	The interpreter does not generate any output.
Any change in the source program after the compilation requires recompiling the entire code.	Any change in the source program during the translation does not require retranslation of the entire code.
Errors are displayed in Compiler after Compiling together at the current time.	Errors are displayed in every single line.

System Development

- **System Development Life Cycle**
- Waterfall Model – Design
- Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project. In "The Waterfall" approach, the whole process of software development is divided into separate phases. In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially.
- The following illustration is a representation of the different phases of the Waterfall Model.

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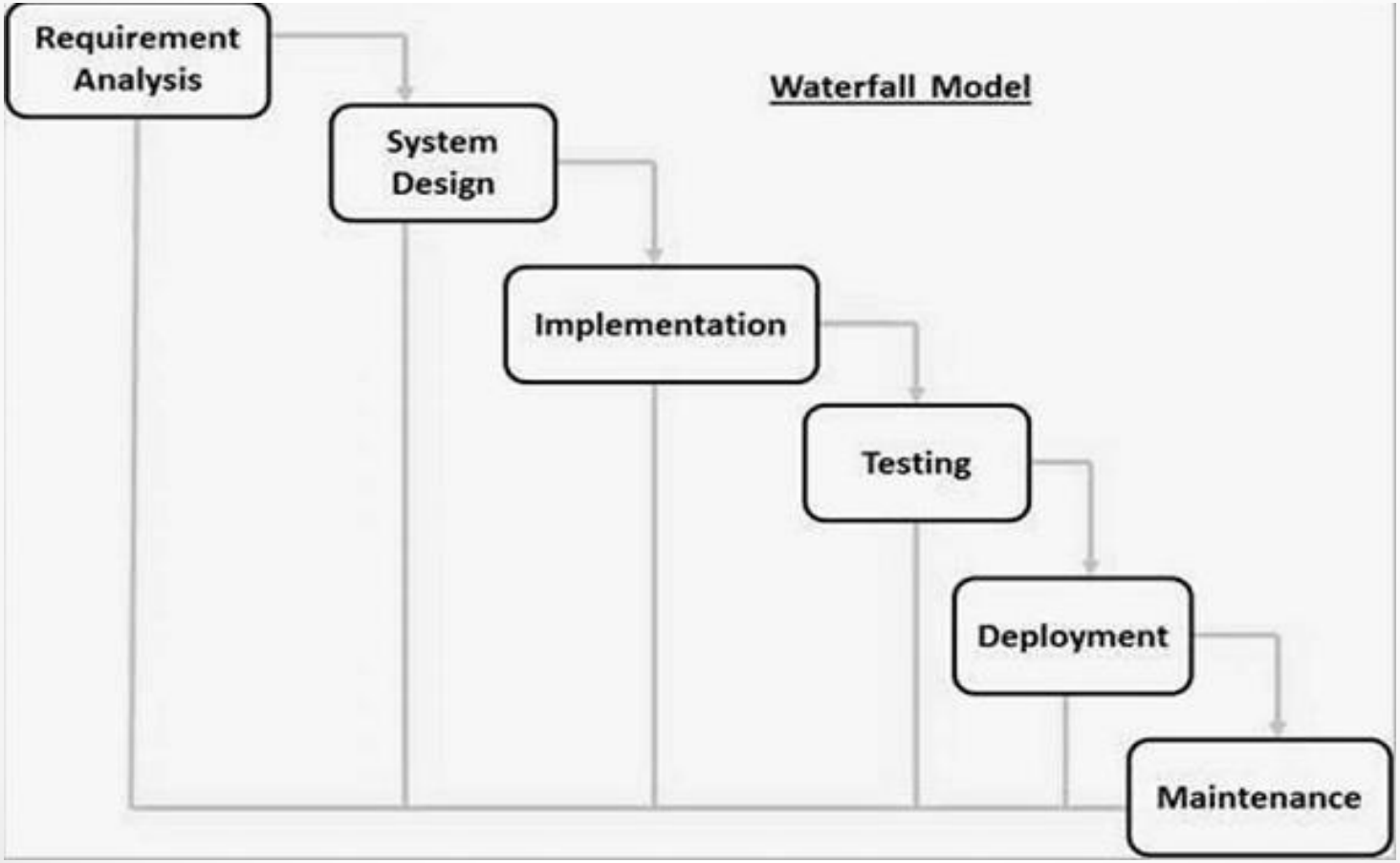

Requirement Analysis
Waterfall Model
System Design
Implementation
Testing
Deployment
Maintenance

Contd..

- **Requirement Gathering and analysis** − All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification document.
- **System Design** − The requirement specifications from first phase are studied in this phase and the system design is prepared. This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.
- **Implementation** − With inputs from the system design, the system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality, which is referred to as Unit Testing.
- **Integration and Testing** − All the units developed in the implementation phase are integrated into a system after testing of each unit. Post integration the entire system is tested for any faults and failures.
- **Deployment of system** − Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.
- **Maintenance** − There are some issues which come up in the client environment. To fix those issues, patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

Program Development &Testing

- Understanding the problem, Develop a solution, Write the Program and then test it.

Testing Contains 2 types

- Black Box Testing: Testing the program without knowing how it works and what is inside it
- White Box Testing: Assumes that Tester knows everything about the program.

Introduction to C Programming

C was first designed by Dennis Ritchie for use with UNIX on DEC PDP-11 computers. The language evolved from Martin Richard's BCPL, and one of its earlier forms was the B language, which was written by Ken Thompson for the DEC PDP-7. The first book on C was The C Programming Language by Brian Kernighan and Dennis Ritchie, published in 1978.

In 1983, the American National Standards Institute (ANSI) established a committee to standardize the definition of C. The resulting standard is known as ANSI C, and it is the recognized standard for the language, grammar, and a core set of libraries. The syntax is slightly different from the original C language, which is frequently called K&R for Kernighan and Ritchie. There is also an ISO (International Standards Organization) standard that is very similar to the ANSI standard.

BASIC STRUCTURE OF C LANGUAGE:

- The program written in C language follows this basic structure. The sequence of sections should be as they are in the basic structure. A C program should have one or more sections but the sequence of sections is to be followed.

1. Documentation section

2.Linking section

3.Definition section

4.Global declaration section

5.Main function section

{

Declaration section Executable section

}

6.Sub program or function section

Contd..

- **DOCUMENTATION SECTION:** comes first and is used to document the use of logic or reasons in your program. It can be used to write the program's objective, developer and logic details. The documentation is done in C language with /* and */ . Whatever is written between these two are called comments.
- **LINKING SECTION :** This section tells the compiler to link the certain occurrences of keywords or functions in your program to the header files specified in this section.
- e.g. #include <stdio.h>

- **DEFINITION SECTION :** It is used to declare some constants and assign them some value.

- e.g. #define MAX 25

- Here #define is a compiler directive which tells the compiler whenever MAX is found in the program replace it with 25.

Contd..

- **GLOBAL DECLARATION SECTION :** Here the variables which are used through out the program (including main and other functions) are declared so as to make them global(i.e accessible to all parts of program)

- e.g. int i; (before main())

- **MAIN FUNCTION SECTION :** It tells the compiler where to start the execution from main()

{

point from execution starts

}

- main function has two sections

declaration section : In this the variables and their data types are declared.

Executable section : This has the part of program which actually performs the task we need.

- **SUB PROGRAM OR FUNCTION SECTION :** This has all the sub programs or the functions which our program needs.

Sample Program

```
#include<stdio.h>
  int main()
                {
                   printf(“Hello”);
                }
```

Output:

Hello

Program for sum of 3 numbers

```
#include<stdio.h>
 int main()
  {
    int a,b,c,sum;
    printf("enter the values:\n");
    scanf("%d%d%d",&a,&b,&c);
     sum=a+b+c;
     printf("sum of numbers =%d",sum);
      return 0;
  }
```

Output:enter values: 2 3 4

Sum of number=9

IDENTIFIERS

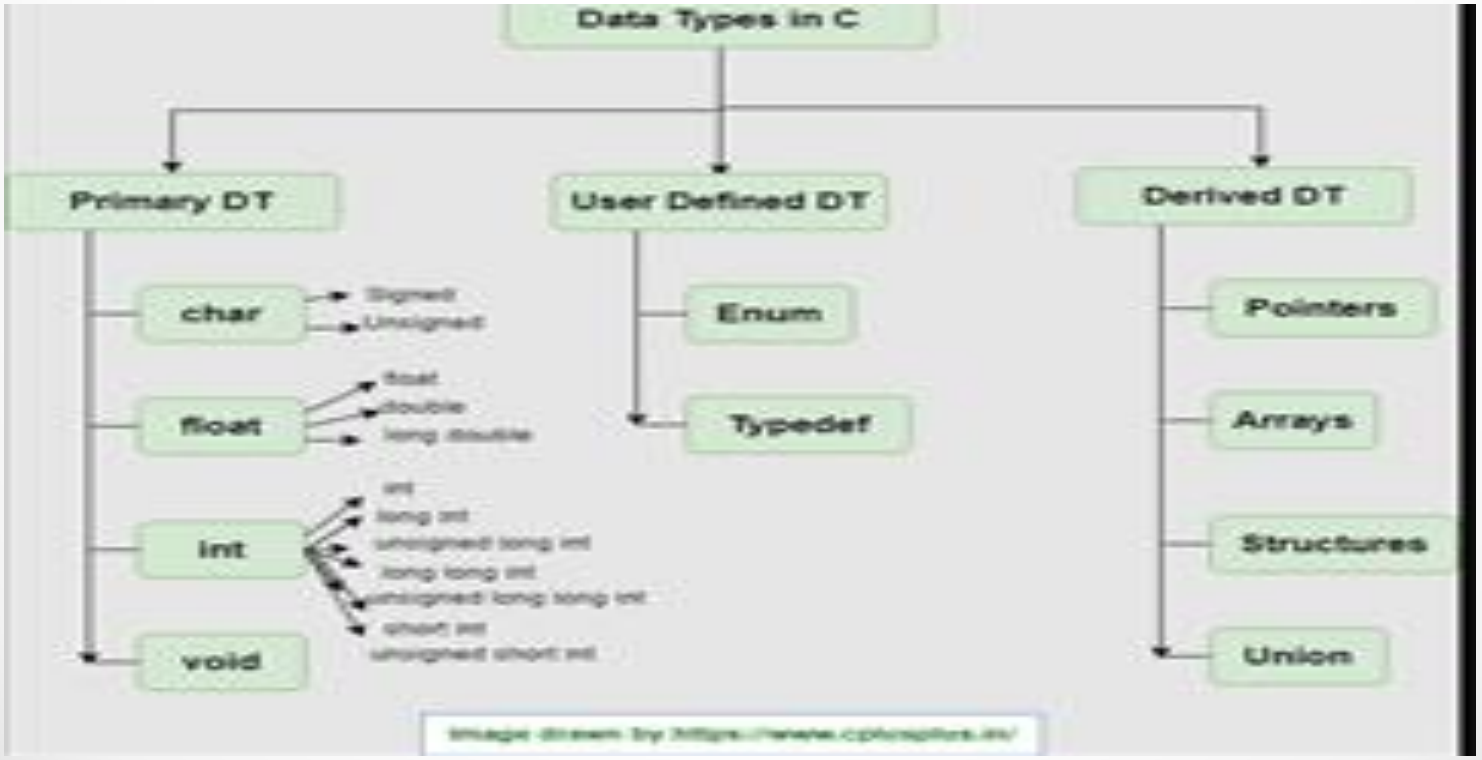
- Names of the variables and other program elements such as functions, array,etc,are known as identifiers.
- There are few rules that govern the way variable are named(identifiers).

- Identifiers can be named from the combination of A-Z, a-z, 0-9, _(Underscore).

- The first alphabet of the identifier should be either an alphabet or an underscore. digit are not allowed.
- It should not be a keyword. Eg: name,ptr,sum

Data Type

- To represent different types of data in C program we need different data types. A data type is essential to identify the storage representation and the type of operations that can be performed on that data. C supports four different classes of data types namely



Example for Integer and float data type

1).Example for Integer data type

```
#include<stdio.h>
int main()
{
  int a,b,sum;
 printf(“enter the values for a & b”);
 scanf(“%d%d”,&a,&b);
sum=a+b;
printf(“sum of a & b=%d”,sum);
return 0;
}
```

2)Example for float data type

```
#include<stdio.h>
int main()
{
  float a,b,sum;
 printf(“enter the values for a & b”);
 scanf(“%f%f”,&a,&b);
sum=a+b;
printf(“sum of a & b=%f”,sum);
return 0;
}
```

Example for double and character

- **Example for Double data type**

```
#include<stdio.h>
int main()
{
  double a,b,sum;
 printf("enter the values for a & b");
 scanf("%lf%lf",&a,&b);
sum=a+b;
printf("sum of a & b=%lf",sum);
return 0;
}
```

Example for Character Data Type:

```
#include<stdio.h>
Int main()
{
char a;
printf(" chracter a="%c",a);
return 0;
}
```

Example of Arrays

```
#include<stdio.h>
int main()
{
  int a[10],n;
 printf(“enter the size of the array”);
 scanf(“%d”,&n);
 printf(“enter the elements”):
for(i=0;i<n;i++){
scanf(“%d”,&a[i]);
printf(“The elements of the array=%d”,a[i]);
return 0;
}
```

How to define Structure/Union

```
struct Student
{
   char name[10];
    int phone;
   int rollno;
    float fees;
};
Stud[10];
```

For union also same way only change struct to union

Typedef And Enum

Example for Enum:

```
#include<stdio.h>
enum week{“sun,mon,tue,wed,thu,fri,sat};
 int main()
{
 enum week today;
 today=wed;
 printf(“Day is %d”,today+1);
}
```

Example for typedef:

```
#include<stdio.h>
Typedef char user[10];
int main()
{
 user  user1=“nithin”;
 return 0;
}
```

Typedef gives an existing data type an alias name.

ASSIGNING VALUES :

- /* write a program to show assigning of values to variables */

```
#include<stdio.h>
main()

{

int a; float b;
printf("Enter any number\n");
b=190.5;
scanf("%d",&a);

printf("user entered %d", a);

printf("B's values is %f", b);

}
```

Formatted Input and Formatted Output

- The simplest of input operator is getchar to read a single character from the input device.

- varname=getchar(); you need to declare varname.

The simplest of output operator is putchar to output a single character on the output device.

putchar(varname)

- The getchar() is used only for one input and is not formatted. Formatted input refers to an input data that has been arranged in a particular format, for that we have scanf.
- scanf("control string", arg1, arg2,...argn);

Contd..

- Control string specifies field format in which data is to be entered.

- arg1, arg2... argn specifies address of location or variable where data is stored.
- eg scanf("%d%d",&a,&b);
- %d used for integers

- %f floats

- %l long

- %c character for formatted output you use printf
- printf("control string", arg1, arg2,...argn);

program to exhibit i/o

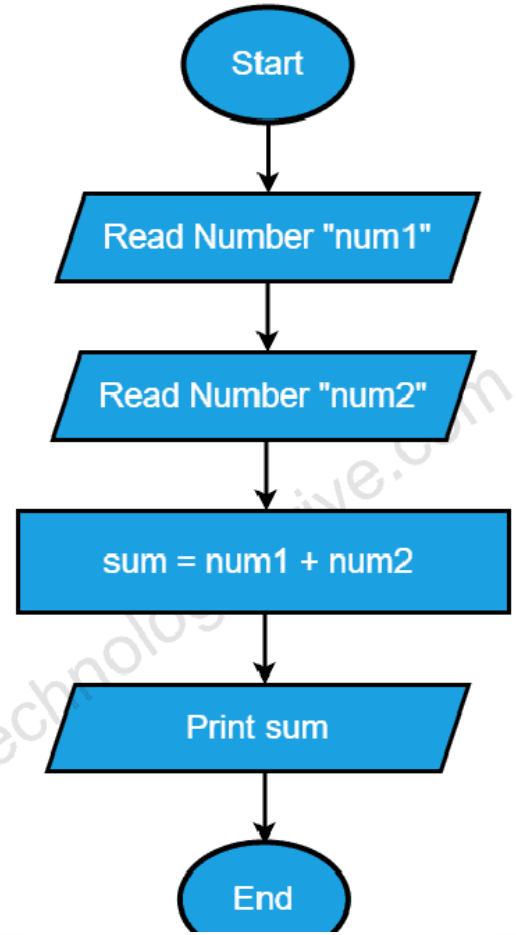
```
#include<stdio.h>

main()

 {

      int a,b; float c;
      printf("Enter any number");
      a=getchar();
      printf("the char is ");
      putchar(a);
      printf("Exhibiting the use of scanf");
     printf("Enter three numbers");
      scanf("%d%d%f",&a,&b,&c);
     printf("%d%d%f",a,b,c);

  }
```

Flowchart & Algorithm

- A Flow chart is a Graphical representation of an Algorithm or a portion of an Algorithm. Flow charts are drawn using certain special purpose symbols such as Rectangles, Diamonds, Ovals and small circles. These symbols are connected by arrows called flow lines.
    - (or)
- The diagrammatic representation of way to solve the given problem is called flow chart.
-
- **The following are the most common symbols used in Drawing flowcharts:**

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectagle represents a process
	Decision	A diamond indicates a decision

Example -Flowchart



Algorithm

- Algorithm is a procedure for solving Mathematical functions
- It is step by step by Step Approach for solving problem.

Example of Algorithm

- STEP 1: Start the program.
- STEP 2: Read the values of ‘a’&’b’.
- STEP 3: Compute the sum of the entered numbers ‘a’,’b’,c=a+b.
- STEP 4: Print the value of ‘c’.
- STEP 5: Stop the program.

Important questions Unit-1

- What is Computer System. Explain computer Hardware with diagram
- List and Explain Different types of Software.
- List out and explain different Programming Languages.
- Explain Basic Structure of C Program with Example
- Explain Different Data types with example
- Explain Formatted input and formatted output
- Write a c program to calculate sum of 3 numbers
- Write a c Program to for simple calculator
- Write a c program for quadratic equation
- Write a program to find area and circumference of the circle.

Case Study

- Profit and loss for small vender shop
- Scenario: Daily income basis

- Thank You

Unit-II

Operators & Expressions

OPERATORS :

- An operator is a symbol that tells the compiler to perform certain mathematical or logical manipulations. They form expressions.
- C operators can be classified as

1.Arithmetic operators

2.Relational operators

3.Logical operators

4.Assignment operators

5.Increment or Decrement operators

6.Conditional operator

7.Bit wise operators

8.Special operators

ARITHMETIC OPERATORS

All basic arithmetic operators are present in C.

operator	meaning
+	add
-	subtract
*	multiplication
/	division
%	modulo division(remainder)

- An arithmetic operation involving only real operands(or integer operands) is called real arithmetic(or integer arithmetic). If a combination of arithmetic and real is called mixed mode arithmetic.

RELATIONAL OPERATORS

- We often compare two quantities and depending on their relation take certain decisions for that comparison we use relational operators.

- Operator Meaning

<	is less than
>	Is greater than
<=	is less than or equal to
>=	is greater than or equal to
==	is equal to
!=	is not equal to

LOGICAL OPERATORS

- An expression of this kind which combines two or more relational expressions is termed as a logical expressions or a compound relational expression. The operators and truth values are

op-1	op-2	op-1 && op-2	op-1 || op-2
non-zero	non-zero	1	1
non-zero	0	0	1
0	non-zero	0	1
0	0	0	0

ASSIGNMENT OPERATORS

- They are used to assign the result of an expression to a variable. The assignment operator is '='.
- v op=exp
- v is variable
- op binary operator
- exp expression
- op= short hand assignment operator

short hand assignment operators

- use of simple assignment operators.

use of short hand assignment operators

- a=a+1 a+=1
- a=a-1 a-=1
- a=a%b a%=b

INCREMENT AND DECREMENT OPERATORS

- ++ and == are called increment and decrement operators used to add or subtract. Both are unary and as follows
- ++m or m++
- --m or m--

- The difference between ++m and m++ is
- if m=5;
  y=++m then it is equal to m=5;

- m++;y=m;
- if m=5;
  y=m++ then it is equal to m=5;
- y=m;m++;

CONDITIONAL OPERATOR

- A ternary operator pair "?:" is available in C to construct conditional expressions of the form

Syntax:

- exp1 ? exp2 : exp3;

It work as

- if exp1 is true then exp2 else exp3

BIT WISE OPERATORS

- C supports special operators known as bit wise operators for manipulation of data at bit level. They are not applied to float or double.

- operator meaning

operator	meaning
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
<<	left shift
>>	right shift
~	one's complement

SPECIAL OPERATORS

- These operators which do not fit in any of the above classification are
- ,(comma), sizeof, Pointer operators(& and *) and member selection operators (. and ->). The comma operator is used to link related expressions together.
- sizeof operator is used to know the sizeof operand.

programs to exhibit the use of operators

```
#include<stdio.h>
int   main()
  {
    int sum, mul, modu;
    float sub, divi;
   int i,j;
   float l, m;
   printf("Enter two integers ");
   scanf("%d%d",&i,&j);
   printf("Enter two real numbers");
  scanf("%f%f",&l,&m);
  sum=i+j; mul=i*j;
  modu=i%j;
  sub=l-m;
  divi=l/m;
  printf("sum is %d", sum);
  printf("mul is %d", mul);
  printf("Remainder is %d", modu);
  printf("subtraction of float is %f", sub);
  printf("division of float is %f", divi);
 return 0;
}
```

program to implement relational and logical

```
#include<stdio.h>
  int main()

  {
      int i, j, k;
     printf("Enter any three numbers ");
    scanf("%d%d%d", &i, &j, &k);
     if((i<j)&&(j<k))
                    printf("k is largest");
     else if i<j || j>k
      {
        if i<j && j >k
        printf("j is largest");
   else
       printf("j is not largest of all");
  }
   }
```

program to implement increment and decrement operators

```
#include<stdio.h>
Int main()
{
 int i;
  printf("Enter a number");
  scanf("%d", &i);
 i++;
 printf("after incrementing %d ", i);
  i--;
 printf("after decrement %d", i);
 return 0;
}
```

program using ternary operator and assignment

```
#include<stdio.h>
int main()

 {

   int i,j,large;

   printf("Enter two numbers ");
   scanf("%d%d",&i,&j);
  large=(i>j)?i:j;

   printf("largest of two is %d",large);

}
```

DECISION MAKING

- We have a number of situations where we may have to change the order of execution of statements based on certain conditions or repeat a group of statements until certain specified conditions are met.

The if statement is a two way decision statement and is used in conjunction with an expression. It takes the following form

If(test expression)

- If the test expression is true then the statement block after if is executed otherwise it is not executed

Syntax:

- if (test expression)

```
{

 statement block;
}
 statement–x ;
```

Example Program

```
#include<stdio.h>
 int  main()
{
  int a,b;
  printf(“Enter two numbers”);
  scanf(“%d%d”,&a,&b):
  if a>b
 printf(“ a is greater”);
 if b>a
printf(“b is greater”);
 return 0;
 }
```

The if –else statement:

- If your have another set of statement to be executed if condition is false then if-else is used

Syntax:

```
if (test expression)
{
  statement block1;
}
else
{
 statement block2;
}
statement –x ;
```

program for if-else

```
#include<stdio.h>
int main()
 {
  int a,b;
 printf(“Enter two numbers”);
 scanf(“%d%d”,&a,&b):
  if a>b
 printf(“ a is greater”)
else
printf(“b is greater”);
return 0;
 }
```

Nesting of if..else statement :

```
•if(text cond1)
        {
            if (test expression2
              {
               statement block1;
               }
      else
              {
                statement block 2;
              }
        }
    else
        {
      statement block2;
      }
     statement-x ;
```

THE SWITCH STATEMENT

- If for suppose we have more than one valid choices to choose from then switch statement in place of if statements.

Syntax:

```
switch(expression)
 {.
 case value-1:
                  block-1
                  break;
case value-2:
                 block-2
                 break;
  --------
default:
                default block;
               break;
}
 statement–x
```

Example Program for Switch

```
#include<stdio.h>
main()
 {
 int marks,index;
char grade[10];
printf(“Enter your marks”);
scanf(“%d”,&marks);
index=marks/10;
 switch(index)
{
 case 10 :
 case 9:
 case 8:
 case 7:
 case 6: grade=”first”;       break;
case 5 : grade=”second”;     break;
case 4 : grade=”third”;       break;
default : grade =”fail”;      break;
}

printf(“%s”,grade);

}
```

Repetition

- **LOOPING :**Some times we require a set of statements to be executed repeatedly until a condition is met.

  We have two types of looping structures.

  1.One in which condition is tested before entering the statement block called entry control.

  2.The other in which condition is checked at exit called exit controlled loop.

WHILE STATEMENT

While(test condition)

{

body of the loop

}

It is an entry controlled loop. The condition is evaluated and if it is true then body of loop is executed. After execution of body the condition is once again evaluated and if is true body is executed once again. This goes on until test condition becomes false

Example program-While

```
#include<stdio.h>
 main()
{

int count,n;
 float x,y;
printf("Enter the values of x and n");

scanf("%f%d",&x,&n);
 y=1.0;
count=1; while(count<=n)
{

y=y*x;

count++;
}

printf("x=%f; n=%d; x to power n = %f",x,n,y);

}
```

DO WHILE STATEMENT

The while loop does not allow body to be executed if test condition is false. The do while is an exit controlled loop and its body is executed at least once.

Syntax:

```
do

{

body

}while(test condition);
```

Example Program-Do While

```
// Program to add numbers until the user enters zero
#include <stdio.h>
int main() {
  double number, sum = 0;
  // the body of the loop is executed at least once
  do {
    printf("Enter a number: ");
    scanf("%lf", &number);
    sum += number;
  }
  while(number != 0.0);
  printf("Sum = %.2lf",sum);
  return 0;
}
```

Output:

Enter a number: 1.5
Enter a number: 2.4
Enter a number: -3.4
Enter a number: 4.2
Enter a number: 0
Sum = 4.70

THE FOR LOOP

It is also an entry control loop that provides a more concise structure

Syntax:

```
 for(initialization; test control; increment)
{

body of loop

}
```

Example –For Loop

```
#include<stdio.h>
main()
{

long int p; int n; double q;
printf(“2 to power n “);
p=1;
for(n=0;n<21;++n)
 {

if(n==0)

p=1;
 else
 p=p*2; q=1.0/(double)p;
printf(“%101d%10d”,p,n);
 }

}
```

Expressions

- An Expression is a sequence of operands and operators.
- A simple Expression contains only one operator.
- Ex:2+3=5
- A complex Expression contains more than one operator.
- Ex:2+5*7

UNCONDITIONAL STATEMENTS

BREAK STATEMENT:

This is a simple statement. It only makes sense if it occurs in the body of a switch, do, while or for statement. When it is executed the control of flow jumps to the statement immediately following the body of the statement containing the break. Its use is widespread in switch statements, where it is more or less essential to get the control .

Example-Break

```
#include <stdio.h>
#include <stdlib.h>
main(){
int i;

for(i = 0; i < 10000; i++){
if(getchar() == 's')
 break;
printf("%d\n", i);
} exit(EXIT_SUCCESS);
}
```

- It reads a single character from the program's input before printing the next in a sequence of numbers.
- If you want to exit from more than one level of loop, the break is the wrong thing to use.

Postfix Expression

- *The expression of the form "a b operator" (ab+) i.e., when a pair of operands is followed by an operator.*
- **Examples:**
- **1)Input**: str = "2 3 1 * + 9 -"
  Output: -4
  Explanation: If the expression is converted into an infix expression, it will be
- 2 + (3 * 1) – 9 = 5 – 9 = -4.
- ***2)Input**: str = "100 200 + 2 / 5 * 7 +"*
  ***Output**: 757*
- *Expalnation:*100 + 200 = 300

  300 / 2 = 150

  150 * 5 = 750

  750 + 7 = 757

Prefix Expression

- To evaluate a postfix expression we can use a stack.
- Iterate the expression from left to right and keep on storing the operands into a stack. Once an operator is received, pop the two topmost elements and evaluate them and push the result in the stack again.
- Prefix and Postfix expressions can be evaluated faster than an infix expression. This is because we don't need to process any brackets or follow operator precedence rule. In postfix and prefix expressions which ever operator comes before will be evaluated first, irrespective of its priority. Also, there are no brackets in these expressions. As long as we can guarantee that a valid prefix or postfix expression is used, it can be evaluated with correctness.

Example

- Input : -+8/632
- 6/3 + 8 - 2
- Output : 8

Precedence	Operator	Description	Associativity
1	++ --	Suffix/postfix increment and decrement	Left-to-right
	()	Function call	
	[]	Array subscripting	
	.	Structure and union member access	
	->	Structure and union member access through pointer	
	(type){list}	Compound literal(C99)	
2	++ --	Prefix increment and decrement[note 1]	Right-to-left
	+ -	Unary plus and minus	
	! ~	Logical NOT and bitwise NOT	
	(type)	Cast	
	*	Indirection (dereference)	
	&	Address-of	
	sizeof	Size-of[note 2]	
	_Alignof	Alignment requirement(C11)	
3	* / %	Multiplication, division, and remainder	Left-to-right
4	+ -	Addition and subtraction	
5	<< >>	Bitwise left shift and right shift	
6	< <=	For relational operators < and ≤ respectively	
	> >=	For relational operators > and ≥ respectively	
7	== !=	For relational = and ≠ respectively	
8	&	Bitwise AND	
9	^	Bitwise XOR (exclusive or)	
10	|	Bitwise OR (inclusive or)	
11	&&	Logical AND	
12	||	Logical OR	
13	?:	Ternary conditional[note 3]	Right-to-left
14	=	Simple assignment	

Evaluate Expression

- 1.)x=3*4+5*6
- First, we perform the multiplications: 3×4=12
- 3×4=12
- 5×6=30
- 5×6=30
- Next, we perform the additions: 12+30=42 12+30=42
- So, ⍰=42*x*=42.

- 2)x=3*(4+5)*6
- First, we perform the operation inside the parentheses: 4+5=94+5=9
- Next, we multiply: 3×9=273×9=27
- Finally, we multiply by 6: 27×6=16227×6=162
- So, ⍰=162*x*=162.

- 3)x=3*4%5/2
- 4)x=3*(4%5)/2

- 5)x=3*((4%5)/2)
- Calculate 4mod 54mod5, which is the remainder of the division of 4 by 5: 4mod 5=44mod5=4
- Divide the result by 2: 4/2=24/2=2
- Multiply the result by 3: 3×2=63×2=6
- So, ⍰=6*x*=6.

- 6)x=17-8/4*2+3-++a (assume a=6)

For given values int a=0,b=1,c=-1
float x=2.5,y=0.0 Evaluate following expressions

- **1)a+=b-=c*10**

- Ans evaluates to −1×10=−10−1×10=−10
- *b*−=*c*×10 is equivalent to *b*=*b*−(−10), which simplifies to *b*=*b*+10. Since *b* was 1, it becomes 1+10=11*b*=1+10=11.

  +=*b* is equivalent to *a*=*a*+*b*, so 0+11=11 therefore after evaluation a=11,b=11mc=-1

- **2)-a*(5+b)/2-c++*b**

a evaluates to 00 because a is 0

5+*b* evaluates to 5+1=65+1=6.

−*a*×(5+*b*) evaluates to 0×6=00×6=0.

0/20/2 evaluates to 00.

c++ increments *c* by 11 (post-increment), so *c* becomes 00 after this operation.

c×*b* evaluates to 0×1=00×1=0.

Subtracting 00 from 00 (from step 4) gives 00.

Therefore, the expression −*a*×(5+*b*)/2−*c*++×*b* evaluates to 00.

Contd..

- **3)a*b*c**

Ans:

0*1*-1=0

- **4)(x>y)+!a||c++**

Ans: int a = 0

int b = 1

int c = -1

float x = 2.5

float y = 0.0

(x > y) evaluates to true since 2.5 is greater than 0.0.

!a evaluates to !0, which is true because ! negates the value.

c++ increments c by 1, so c becomes 0 after this operation.

Now we have true from step 1, true from step 2, and the incremented value of c (which is 0).

true is typically represented as 1 in C-like languages, so we have 1 + 1 + 0.

1 + 1 + 0 evaluates to 2.

Therefore, the expression (x > y) + !a || c++ evaluates to 2 given the provided values

Contd..

- 5)a<b&&c<b

Ans:0<1 && -1<1 true && true=True=1

True=1

- 6)b+c||!a
- Ans:1+(-1)||!1 1-1||0 0||0 =>0
- 7)x*5&&5||(b/c)
- Ans:2.5*5 && 5||(1/-1)
- 12.5 && 5||0
- True||0 -> true

Contd..

- 8)a<=10&&x>=1&&b
- Ans:
- 9)!x||!c||b+c
- Ans:
- 10)x*y<a+b||c
- Ans:2.5*0.0<0+1||-1
- 0<1||-1 ->TRUE||-1
- 1||-1

Type Conversion

- Type conversion in C is the process of converting one data type to another. The type conversion is only performed to those data types where conversion is possible. Type conversion is performed by a compiler. In type conversion, the destination data type can’t be smaller than the source data type. Type conversion is done at compile time and it is also called widening conversion because the destination data type can’t be smaller than the source data type. ***There are two types of Conversion.***
- ***Implicit Type Conversion***
- ***Explicit Type Conversion***

Implicit Type Conversion

- Also known as ‘automatic type conversion’.
- A. Done by the compiler on its own, without any external trigger from the user.
- B. Generally takes place when in an expression more than one data type is present. In such conditions type conversion (type promotion) takes place to avoid loss of data.

Example of Implicit type Conversion

```
#include <stdio.h>
int main()
{
    int x = 10; // integer x
    char y = 'a'; // character c

    // y implicitly converted to int. ASCII
    // value of 'a' is 97
    x = x + y;

    // x is implicitly converted to float
    float z = x + 1.0;

    printf("x = %d, z = %f", x, z);
    return 0;
}
```

Output:

x = 107, z = 108.000000

Explicit Type Conversion

- This process is also called type casting and it is user-defined. Here the user can typecast the result to make it of a particular data type. The syntax in C Programming:

Syntax:

- (type) expression

Type indicated the data type to which the final result is converted.

Example of Explicit Type conversion

```
#include<stdio.h>

int main()
{
    double x = 1.2;

    // Explicit conversion from double to int
    int sum = (int)x + 1;

    printf("sum = %d", sum);

    return 0;
}
```

Output: sum = 2

CONTINUE STATEMENT:

- This statement has only a limited number of uses. The rules for its use are the same as for break, with the exception that it doesn't apply to switch statements. Executing a continue starts the next iteration of the smallest enclosing do, while or for statement immediately. The use of continue is largely restricted to the top of loops, where a decision has to be made whether or not to execute the rest of the body of the loop. In this example it ensures that division by zero (which gives undefined behavior) doesn't happen

Example -Continue

```
#include <stdio.h>
 #include <stdlib.h>
main(){
int i;

for(i = -10; i < 10; i++)
{ if(i == 0)
continue;
 printf("%f\n", 15.0/i);
/*
* Lots of other statements .....
*/
} exit(EXIT_SUCCESS);

}
```

Contd..

- The continue can be used in other parts of a loop, too, where it may occasionally help to simplify the logic of the code and improve readability. continue has no special meaning to a switch statement, where break does have. Inside a switch, continue is only valid if there is a loop that encloses the switch, in which case the next iteration of the loop will be started.

GOTO AND LABELS

In C, it is used to escape from multiple nested loops, or to go to an error handling exit at the end of a function.

You will need a *label* when you use a goto;

this example shows both.

```
goto L1;
/* whatever you like here */  L1: /* anything else */
```

A label is an identifier followed by a colon. Labels have their own „name space“ so they can't clash with the names of variables or functions. The name space only exists for the function containing the label, so label names can be re-used in different functions. The label can be used before it is declared, too, simply by mentioning it in a goto statement.

Labels must be part of a full statement, even if it's an empty one. This usually only matters when you're trying to put a label at the end of a compound statement—like this.

```
label_at_end: ; /* empty statement */
}
```

The goto works in an obvious way, jumping to the labelled statements. Because the name of the label is only visible inside its own function, you can't jump from one function to another one.

It's hard to give rigid rules about the use of gotos but, as with the do, continue and the break (except in switch statements), over-use should be avoided. More than one goto every 3–5 functions is a symptom that should be viewed with deep suspicion.

Important Questions

1. List out all operators in C. Write a program using Switch include all operators.
2. Analyze the following code write the output.

```
#include<stdio.h>
void main()
{
printf("7 && 0=%d\n",7&&0);
printf("7 && 0=%d\n",7||0);
printf("!0=%d\n",!0);
}
```

3. What are Infix, Prefix, Postfix Expression.

Solve the following prefix and Postfix Expressions. For given values int a=0,b=1,c=-1 float x=2.5,y=0.0 Evaluate following expressions.

1.(x>y) +! a||c++

2. X*5&&5||(b/c)

3.!x||!c||b+c

4.a<=10&&x>=1&&b

5.-a*(5+b)/2-c++*b

6.x*y<a+b||c

4.What are Implicit and Explicit Type Conversion? Explain with example.

5.List out and explain the different types of Loops in C

6.Write a C Program to print a given integer number is palindrome or not

7.Discuss break and continue statements with suitable examples.

8.Program to calculate the total sales given the unit price, quantity, discount and tax rate

9.Write a C-Program to calculate a student's average score for a course with 4 quizzes, 2 midterms and a final. The quizzes are weighted 30%, the midterms 40% and the final 30%.

10.

11.Write C-Program to evaluate two complex expressions.

12.Write C-Program to read a test score, calculate the grade for the score and print the grade

13.Write a C Program to print Square of Index and Print it.

Case study

Case Study: To display Students grade

Scenario: Only Student Grade List of Ist year students of Cyber Security Branch

Case Study:Set Password

Scenario:After Setting Password for wrong Entry

Ex:0 and o

Unit-III

Arrays

Arrays: Concepts, Using arrays in C:

- An array is a group of related data items that share a common name.
- Ex:- Students
- The complete set of students are represented using an array name students.
- A particular value is indicated by writing a number called index number or subscript in brackets after array name. The complete set of value is referred to as an array, the individual values are called elements.

Accessing the array elements

- **Accessing the array elements:** Accessing the array element is done using a loop statement in combination with printf statements or any other processing statements. Example of accessing the array elements and calculating total marks is given below:

Example:

```
#include<stdio.h>
int main( )
{
int total=0, marks[4]={35,44,55,67};
for(i=0; i<4; i++)
total=total+marks[i]; /* calculating total marks*/
 printf("Total marks=%d",total);
 return 0;
}
```

Output: Total marks=201

Write a program to search an element using Binary Search

```
#include <stdio.h>
int main()
{
int i, low, high, mid, n, key, array[100];
printf("Enter number of elements:");
scanf("%d",&n);
printf("Enter %d integers", n);
for(i = 0; i < n; i++)
scanf("%d",&array[i]);
printf("Enter value to find");
scanf("%d", &key);
low = 0;
high = n - 1;
mid = (low+high)/2;
while (low <= high) {
if(array[mid] < key)
low = mid + 1;
else if (array[mid] == key) {
printf("%d found at location %d", key, mid+1);
break;
}
else
high = mid - 1;
mid = (low + high)/2;
}
if(low > high)
printf("Not found! %d isn't present in the list", key);
return 0;
}
```

Output:

```
 Enter number of elements:3
Enter 3 integers 1 2 3
1 2 3
Enter value to find 3
3
3 found at location 3
```

Write Program to sort numbers using Bubble Sort

```
#include <stdio.h>
int main()
{
int array[100], n, i, j, swap;
printf("Enter number of elements\n");
scanf("%d", &n);
printf("Enter %d integers\n", n);
for (i = 0; i< n; i++)
scanf("%d", &array[i]);
for (i = 0 ;i< n - 1; i++)
{for (j = 0 ;j< n - i - 1; j++){
if (array[j] > array[j+1])
{swap = array[j];
array[j] = array[j+1];
array[j+1] = swap;}}}
printf("Sorted list in ascending order:\n");
for (i = 0; i< n; i++)
printf("%d\n", array[i]);
return 0;
}
```

Output:

```
Enter number of elements
3
Enter 3 integers
5 2  1
Sorted list in ascending order:
1
2
5
```

ONE – DIMENSIONAL ARRAYS :

- A list of items can be given one variable index is called single subscripted variable or a one-dimensional array.
- The subscript value starts from 0. If we want 5 elements the declaration will be int number[5];
- The elements will be number[0], number[1], number[2], number[3], number[4] There will not be number[5]

Declaration of One - Dimensional Arrays :

Type variable – name [sizes];

Type – data type of all elements Ex: int, float etc., Variable – name – is an identifier

Size – is the maximum no of elements that can be stored. Ex:- float avg[50]

This array is of type float. Its name is avg. and it can contains 50 elements only. The range starting from 0 – 49 elements.

Initialization of Arrays :

Initialization of elements of arrays can be done in same way as ordinary variables are done when they are declared.

Type array name[size] = {List of Value};

Ex:- int number[3]={1,2,3};

Program Showing one dimensional array

```
#include<stdio.h>

main()
 {

int i;
 float x[10],value,total;

printf("Enter 10 real numbers\n");
 for(i=0;i<10;i++)
{

scanf("%f",&value);
x[i]=value;
}
 total=0; for(i=0;i<10;i++)
total=total+x[i] ;
for(i=0;i<10;i++)
printf("x*%2d+=%5.2f\n",I+1,x*I+);
 printf("total=%0.2f",total);
}
```

TWO – DIMENSIONAL ARRAYS:

- To store tables we need two dimensional arrays. Each table consists of rows and columns. Two dimensional arrays are declare as type array name [row-size][col-size];
- Ex:int a[10][10];

INITIALIZING TWO DIMENSIONAL ARRAYS:

They can be initialized by following their declaration with a list of initial values enclosed in braces.

Ex:- int table[2][3] = {0,0,0,1,1,1};

Initializes the elements of first row to zero and second row to one. The initialization is done by row by row. The above statement can be written as

int table[2][3] = {{0,0,0},{1,1,1}};

When all elements are to be initialized to zero, following short-cut method may be used.

int m[3][5] = {{0},{0},{0}};

•

Program to find Transpose of a Matrix

```
#include <stdio.h>
int main()
{
int m, n, c, d, matrix[10][10], transpose[10][10];
printf("Enter the number of rows and columns of a matrix\n");
scanf("%d%d", &m, &n);
printf("Enter elements of the matrix\n");
for (c = 0; c < m; c++)
for (d = 0; d < n; d++)
scanf("%d", &matrix[c][d]);
for (c = 0; c < m; c++)
for (d = 0; d < n; d++)
transpose[d][c] = matrix[c][d];
printf("Transpose of the matrix:\n");
for (c = 0; c < n; c++) {
for (d = 0; d < m; d++)
printf("%d\t", transpose[c][d]);
printf("\n");
}
return 0;
}
```

Output:

Enter the number of rows and columns of a matrix

2 2

Enter elements of the matrix

2 3 4 5

Transpose of the matrix:

2 4

3 5

Applications of Array

- Data elements can be sorted using arrays. Arrays are used to quickly store and sort elements in a variety of sorting techniques, including Bubble sort, Insertion sort, and Selection sort.
- Matrix operations can be done with arrays. Numerous databases, both small and large, are made up of records-based one- and two-dimensional arrays.

Advantages & Disadvantages of Array

Advantages	Disadvantages
This is a practical method of storing a specified number of similar-type items.	The array can only store a predetermined amount of elements in accordance with the initial size specification. There is no mechanism for expanding the array's size to include additional data down the road, should that be necessary.
Data memory allocation is carried out progressively without using additional memory.	The excess memory space that has already been allocated does not get used if the actual number of elements is fewer than the specified size of an array.
It is quicker to access the arrays by their index value.	Utilizing only one type of data might occasionally have restrictions since, in real-world situations, it may be necessary to use many types of data in array form.
It enables the multidimensional array storage of matrix data items.	It is inconvenient and time-consuming to add or remove records from the array since we must manage memory and indexing.

Functions in C

- A function is a self contained program segment that carries out some specific well defined tasks.
- **Advantages of functions:**
- 1. Write your code as collections of small functions to make your program modular 2.Structured programming
- Code easier to debug 4.Easier modification 5.Reusable in other programs

Function Definition

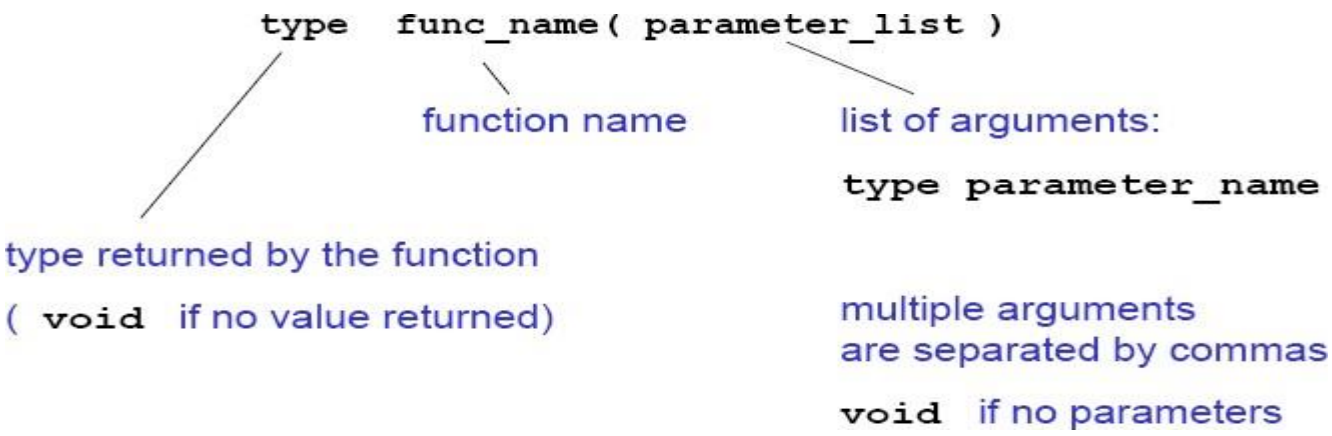
- **Function Definition :**

type func_name(parameter list)

{

declarations; statements;
}

FUNCTION HEADER :



Examples:

```
int fact(int n)

void error_message(int errorcode)

double initial_value(void)

int main(void)
```

Usage:

```
a = fact(13);

error_message(2);

x=initial_value();
```

CATEGORIES OF FUNCTIONS:

- A function depending on whether arguments are present or not and whether a value is returned or not may belong to.
- Functions with no arguments and no return values.
- Functions with arguments and no return values.
- Functions with arguments and return values.

- **Functions with no arguments and no return values :**
- When a function has no arguments, it does not receive any data from calling function.
- In effect, there is no data transfer between calling function and called function.

Arguments but no return values:

- The function takes argument but does not return value.

- The actual (sent through main) and formal(declared in header section) should match in number, type and order.
- In case actual arguments are more than formal arguments, extra actual arguments are discarded. On other hand unmatched formal arguments are initialized to some garbage values.

user-defined function

- A **user-defined function** is a type of function in C language that is defined by the user himself to perform some specific task. It provides code reusability and modularity to our program. User-defined functions are different from built-in functions as their working is specified by the user and no header file is required for their usage.
- Syntax:

return_type **function_name** (type1 *arg1*, type2 *arg2* .... typeN *argN*)

{ *// actual statements to be executed // return value if any*

}

Facctorial of number using Function

```
#include<stdio.h>
#include<math.h>
int main()
{

   printf("Enter a Number to Find Factorial: ");
   printf("\nFactorial of a Given Number is: %d ",fact());
   return 0;
}
int fact()
{
   int i,fact=1,n;
   scanf("%d",&n);
   for(i=1; i<=n; i++)
   {
      fact=fact*i;
   }
   return fact;
}
```

Output:

Enter a Number to Find Factorial: 4

Factorial of a Given Number is: 24

PARAMETER PASSING TECHNIQUES:

- Parameter passing mechanism in C is of two types.
  - Call by Value
  - Call by Reference.

- The process of passing the actual value of variables is known as Call by Value.
- The process of calling a function using pointers to pass the addresses of variables is known as Call by Reference. The function which is called by reference can change the value of the variable used in the call.

Example of Call by Value:

```
#include <stdio.h>
void swap(int,int);
main()
{
int a,b;
printf("Enter the Values of a and b:");
 scanf("%d%d",&a,&b);
 printf("Before Swapping \n"); printf("a = %d \t b = %d", a,b);
swap(a,b);
printf("After Swapping \n"); printf("a = %d \t b = %d", a,b);
}

void swap(int a, int b)
{
int temp; temp = a; a = b;
b = temp;
}
```

Example of Call by Reference:

```
#include<stdio.h>
main()
{
int a,b; a = 10;
b = 20;
swap (&a, &b); printf(“After Swapping \n”);
printf(“a = %d \t b = %d”, a,b);
}
void swap(int *x, int *y)
{
int temp; temp = *x;
*x = *y;
*y = temp;
}
```

RECURSIVE FUNCTIONS:

Recursion is a repetitive process in which a function calls itself (or) A function is called recursive if it calls itself either directly or indirectly. In C, all functions can be used recursively.

Example: Fibonacci Numbers

A recursive function for Fibonacci numbers (0,1,1,2,3,5,8,13...)

```
/* Function with recursion*/

int fibonacci(int n)

{

if (n <= 1) return n; else
return (fibonacci(n-1) + fibonacci(n-2));

}
```

With recursion 1.4×10^9 function calls needed to find the 43rd Fibonacci number(which has the value 433494437) .If possible, it is better to write iterative functions.

```
int factorial (int n) /* iterative version */

{

for ( ; n > 1; --n) product *= n; return product;
}
```

Factorial of a number using Recursion

```
#include<stdio.h>

int fact(int);
int main()
{
   int x,n;
   printf(" Enter the Number to Find Factorial :");
   scanf("%d",&n);

   x=fact(n);
   printf(" Factorial of %d is %d",n,x);

   return 0;
}
int fact(int n)
{
   if(n==0)
      return(1);
   return(n*fact(n-1));
}
```

- **Fibonacci Series using Recursion**

```
#include<stdio.h>
 int Fibonacci(int);
 int main()
{
   int n, i = 0, c;
    scanf("%d",&n);
    printf("Fibonacci series\n");
    for ( c = 1 ; c <= n ; c++ )
   {
      printf("%d\n", Fibonacci(i));
      i++;
   }

   return 0;
}
 int Fibonacci(int n)
{
   if ( n == 0 )
      return 0;
   else if ( n == 1 )
      return 1;
   else
      return ( Fibonacci(n-1) + Fibonacci(n-2) );
}
```

Unit IV

Storage Classes, Pointers

STORAGE CLASSES:

- In ‘C’ a variable can have any one of four Storage Classes.

- Automatic Variables
- External Variables
- Static Variables
- Register Variables

AUTOMATIC VARIABLES :

They are declared inside a function in which they are to be utilized. They are created when function is called and destroyed automatically when the function is exited, hence the name automatic. Automatic Variables are therefore private (or local) to the function in which they are declared. Because of this property, automatic variables are also referred to as local or internal variables.

By default declaration is automatic. One important feature of automatic variables is that their value cannot be changed accidentally by what happens in some other function in the program.

Contd..

```
#include<stdio.h>
 main()
{
 int m=1000;
func2();
printf(“%d\n”,m);

}
  func1()
 {
 int m=10;
 printf(“%d\n”,m);
 }
func2()
 {
 int m=100;
func1();
printf(“%d”,m);

}
```

Contd..

- First, any variable local to main will normally live throughout the whole program, although it is active only in main.
- Secondly, during recursion, nested variables are unique auto variables, a situation similar to function nested auto variables with identical names.

EXTERNAL VARIABLES :

- Variables that are both alive and active throughout entire program are known as external variables. They are also known as Global Variables. In case a local and global have same name local variable will have precedence over global one in function where it is declared.

```
#include<stdio.h>
int x;
main()
 {
 x=10;
 printf(“%d”,x);

printf(“x=%d”,fun1());

printf(“x=%d”,fun2());

printf(“x=%d”,fun3());
  }
```

```
 fun1()

{

x=x+10;

return(x);

}

fun2()

{

int x; x=1;
return(x);

}

fun3()

{ x=x+10;
return(x);

}
```

An extern within a function provides the type information to just that one function.

STATIC VARIABLES:

- The value of Static Variable persists until the end of program. A variable can be declared Static using Keyword Static like Internal & External Static Variables are differentiated depending whether they are declared inside or outside of auto variables, except that they remain alive throughout the remainder of program.

```
#include<stdio.h>
main()
{

int I;

for (I=1;I<=3;I++)

stat();

}

stat()

{

static int x=0; x=x+1;
printf("x=%d\n",x);

}
```

REGISTER VARIABLES:

- We can tell the Compiler that a variable should be kept in one of the machines registers, instead of keeping in the memory. Since a register access is much faster than a memory access, keeping frequently accessed variables in register will lead to faster execution
- Syntax:
- register int Count

Strings

- **STRINGS (CHARACTER ARRAYS) :**
- A String is an array of characters. Any group of characters (except double quote sign)defined between double quotes is a constant string. Ex: “C is a great programming language”.
- If we want to include double quotes.

- Ex: “\”C is great \” is norm of programmers “.

Declaring and initializing strings :-

- A string variable is any valid C variable name and is always declared as an array.

- char string name [size];
- size determines number of characters in the string name. When the compiler assigns a character string to a character array, it automatically supplies a null character (‘\0’) at end of String. Therefore, size should be equal to maximum number of character in String plus one.
- String can be initialized when declared as
- 1. char city*10+= “NEW YORK’;
- 2. char city[10]= ,‘N’,’E’,’W’,’ ‘,’Y’,’O’,’R’,’K’,’/0’-;

- C also permits us to initializing a String without specifying size.

- Ex:- char Strings* += ,‘G’,’O’,’O’,’D’,’\0’-;

READING STRINGS FROM USER

- %s format with scanf can be used for reading String. char address[15];
- scanf(“%s”,address);
- The problem with scanf function is that it terminates its input on first white space it finds. So scanf works fine as long as there are no spaces in between the text.

Reading a line of text:

- If we are required to read a line of text we use getchar(). which reads a single characters. Repeatedly to read successive single characters from input and place in character array.

Program to read String using scanf & getchar

```
#include<stdio.h>
main()
{
 char line[80],ano_line[80],character;
int c;
c=0;

printf(“Enter String using scanf to read \n”);
scanf(“%s”, line);
printf(“Using getchar enter new line\n”);
 do
{

character = getchar();
ano_line[c] = character;
c++;
-     while(character !=’\n’);
-     c=c-1;
ano_line*c+=’\0’;
}
```

STRING HANDLING/MANIPULATION FUNCTIONS:

- strcat( ) Concatenates two Strings
- strcmp( ) Compares two Strings
- strcpy( ) Copies one String Over another
- Strlen() Finds length of String

Contd..

- **strcat() function:**
- This function adds two strings together.
- Syntax:
- char *strcat(const char *string1, char *string2);
- strcat(string1,string2); string1 = VERY

- string2 = FOOLISH

- strcat(string1,string2);

- string1=VERY FOOLISH string2 = FOOLISH

Strncat:.

- **Strncat :**Append n characters from string2 to stringl.
- char *strncat(const char *string1, char *string2, size_t n);
- **strcmp() function :**
- This function compares two strings identified by arguments and has a value 0 if they are equal. If they are not, it has the numeric difference between the first non-matching characters in the Strings.

Contd..

- Syntax: int strcmp (const char *string1,const char *string2);
- strcmp(string1,string2);
- Ex:- strcmp(name1,name2); strcmp(name1,”John”); strcmp(“ROM”,”Ram”);
- **Strncmp:** Compare first n characters of two strings.

strcpy() function

- It works almost as a string assignment operators. It takes the form
- Syntax: char *strcpy(const char *string1,const char *string2);
- strcpy(string1,string2);
- string2 can be array or a constant.
- **Strncpy:** Copy first n characters of string2 to stringl .
- char *strncpy(const char *string1,const char *string2, size_t n);

strlen() function

- **strlen() function :**
- Counts and returns the number of characters in a string.
- Syntax:int strlen(const char *string);
- n= strlen(string);
- n-> integer integer variable which receives the value of length of string.

Illustration of string-handling

```
#include<stdio.h>
#include<string.h>
main()
{
 char s1[20],s2[20],s3[20];
int X,L1,L2,L3;
printf(“Enter two string constants\n”);
 scanf(“%s %s”,s1,s2);
X=strcmp(s1,s2);
if (X!=0)
{
 printf(“Strings are not equal\n”);
strcat(s1,s2);
}
 else
 printf(“Strings are equal \n”);
 strcpy(s3,s1);
L1=strlen(s1);
  L2=strlen(s2);
 L3=strlen(s3);
 printf(“s1=%s\t length=%d chars \n”,s1,L1);
printf(“s2=%s\t length=%d chars \n”,s2,L2);
printf(“s3=%s\t length=%d chars \n”,s3,L3);
}
```

Pointers

- One of the powerful features of C is ability to access the memory variables by their memory address. This can be done by using Pointers. The real power of C lies in the proper use of Pointers.
- A pointer is a variable that can store an address of a variable (i.e., 112300).We say that a pointer points to a variable that is stored at that address. A pointer itself usually occupies 4 bytes of memory (then it can address cells from 0 to 232-1).

Advantages of Pointers:

- A pointer enables us to access a variable that is defined out side the function.
- Pointers are more efficient in handling the data tables.
- Pointers reduce the length and complexity of a program.
- They increase the execution speed.

Definition :

A variable that holds a physical memory address is called a pointer variable or Pointer.

Declaration :

Datatype * Variable-name;

Eg:- int *ad; /* pointer to int */

char *s; /* pointer to char */

float *fp; /* pointer to float */

char **s; /* pointer to variable that is a pointer to char */

A pointer is a variable that contains an address which is a location of another variable in memory.

Consider the Statement
p=&i;

- Here “&‟ is called address of a variable.
- “p‟ contains the address of a variable i.

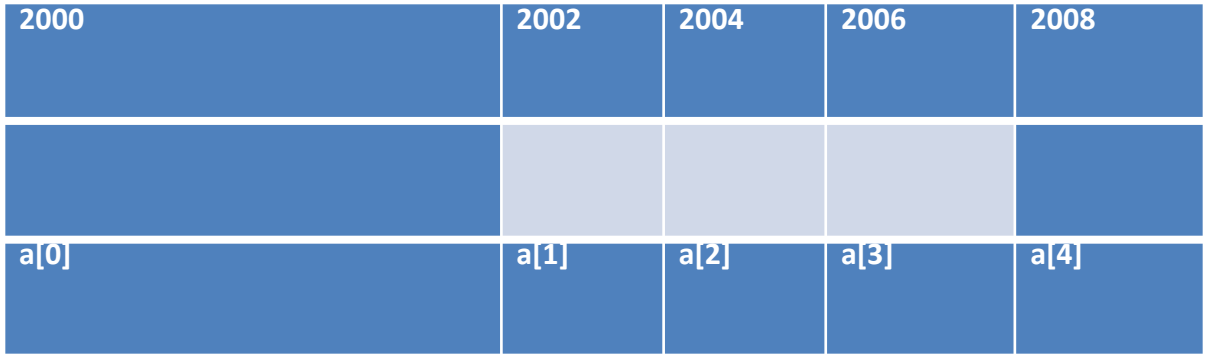
- The operator & returns the memory address of variable on which it is operated, this is called Referencing.
- The * operator is called an indirection operator or dereferencing operator which is used to display the contents of the Pointer Variable.

- Consider the following Statements :

- int *p,x; x =5;
- p= &x;

POINTERS WITH ARRAYS :

- When an array is declared, elements of array are stored in contiguous locations. The address of the first element of an array is called its base address.
- Consider the array



- The name of the array is called its base address. i.e., a and k& a[20] are equal.
- Now both a and a[0] points to location 2000. If we declare p as an integer pointer, then we can make the pointer P to point to the array a by following assignment
- P = a;

- We can access every value of array a by moving P from one element to another. i.e., P points to 0^{th} element
- P+1 points to 1^{st} element
- P+2 points to 2^{nd} element
- P+3 points to 3^{rd} element
- P +4 points to 4^{th} element

Reading and Printing an array using Pointers :

```
#include<stdio.h>
main()
{

 int *a,i;
printf("Enter five elements:");
for (i=0;i<5;i++)
scanf("%d",a+i);
printf("The array elements are:");
 for (i=0;i<5;i++)
printf("%d", *(a+i));



}
```

In one dimensional array, a[i] element is referred by
(a+i) is the address of i^{th} element.
* (a+i) is the value at the i^{th} element.

In two-dimensional array, a[i][j] element is represented as
((a+i)+j)

Unit V

Derived Types, Structures and Unions, File Handling

Derived Types

- Enumerated Types: Declaring an Enumerated Type, Operations on Enumerated Types, Initializing Enumerated Constants.
- Enumeration or Enum in C is a special kind of data type defined by the user. It consists of constant integrals or integers that are given names by a user. The use of enum in C to name the integer values makes the entire program easy to learn, understand, and maintain by the same or even different programmer.

Syntax of Enum

An enum is defined by using the ‘enum’ keyword in C, and the use of a comma separates the constants within.

The basic syntax of defining an enum is:

enum enum_name{int_const1, int_const2, int_const3, …. int_constN};

In the above syntax, the default value of int_const1 is 0, int_const2 is 1, int_const3 is 2, and so on. However, you can also change these default values while declaring the enum. Below is an example of an enum named cars and how you can change the default values.

enum cars{BMW, Ferrari, Jeep, Mercedes-Benz};

Here, the default values for the constants are:

BMW=0, Ferrari=1, Jeep=2, and Mercedes-Benz=3.

However, to change the default values, you can define the enum as follows:

```
enum cars{
BMW=3,
Ferrari=5,
Jeep=0,
Mercedes-Benz=1
};
```

Enumerated Type Declaration to Create a Variable

Similar to pre-defined data types like int and char, you can also declare a variable for enum and other user-defined data types.

Here's how to create a variable for enum.

enum condition (true, false); //declaring the enum

enum condition e; //creating a variable of type condition

Suppose we have declared an enum type named condition; we can create a variable for that data type as mentioned above. We can also converge both the statements and write them as:

enum condition (true, false) e;

For the above statement, the default value for true will be 1, and that for false will be 0.

How to Create and Implement enum

Now that we know how to define an enum and create variables for it, let's look at some examples to understand how to implement enum in C programs.

Printing the Values of Weekdays

```
#include <stdio.h>
    enum days{Sunday=1, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday};
int main(){
    // printing the values of weekdays
    for(int i=Sunday;i<=Saturday;i++){
        printf("%d, ",i);
    }
    return 0;
}
```

Output:

1,2,3,4,5,6,7,

In the above code, we declared an enum named days consisting of the name of the weekdays starting from Sunday. We then initialized the value of Sunday to be 1. This will assign the value for the other days as the previous value plus 1. To iterate through the enum and print the values of each day, we have created a for loop and initialized the value for i as Sunday.

STRUCTURES :

- A Structure is a collection of elements of dissimilar data types. Structures provide the ability to create user defined data types and also to represent real world data.
- Suppose if we want to store the information about a book, we need to store its name (String), its price (float) and number of pages in it(int). We have to store the above three items as a group then we can use a structure variable which collectively store the information as a book.
- Structures can be used to store the real world data like employee, student, person etc.

Contd..

Declaration :

The declaration of the Structure starts with a Key Word called Struct and ends with ; .
The Structure elements can be any built in types.

struct <Structure name>
{
Structure element 1;
Structure element 2;
-
-
-
Structure element n;
};

Then the Structure Variables are declared as
struct < Structure name > <Var1,Var2>;
Eg:-
struct emp
{
int empno.
char ename[20]; float sal;
};
struct emp e1,e2,e3;

- The Structure can also be declared as :

```
struct emp
{
        int empno;
        char ename[20];
        float sal;
}e1,e2,e3;
```

(or)

```
struct
{
int empno;
char ename[20];
float sal;
}e1,e2,e3;
```

Initialization :

Initialization :

Structure Variables can also be initialised where they are declared. struct emp

{

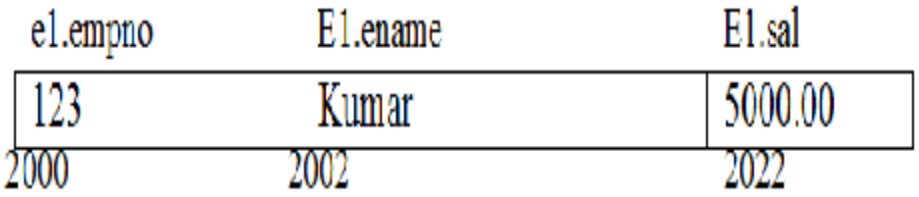
int empno;

char ename[20]; float sal;

};

struct emp e1 = { 123,”Kumar”,5000.00};

- To access the Structure elements we use the .(dot) operator. To refer empno we should use e1.empno
- To refer sal we whould use e1.sal
- Structure elements are stored in contiguous memory locations as shown below. The above Structure occupies totally 26 bytes.



Program to illustrate the usage of a Structure.

```
#include<stdio.h>
main()
{
struct emp
{
int empno;
char ename[20];
float sal;
};
struct emp e;
printf (“ Enter Employee number: \n”);
scanf(“%d”,&e.empno);
printf (“ Enter Employee Name: \n”);
scanf(“%s”,&e.empname);
printf (“ Enter the Salary: \n”);
 scanf(“%f”,&e.sal);
printf (“ Employee No = %d”, e.empno);
 printf (“\n Emp Name = %s”, e.empname);
printf (“\n Salary = %f”, e.sal);
}
```

Program to read Student Details and Calculate total and average using structures

```
#include<stdio.h> main()
{
struct stud
{
int rno;
char sname[20];

int m1,m2,m3;
};
struct stud s;
int tot;
float avg;
 printf(“Enter the student roll number: \n”);
scanf(“%d”,&s.rno);
printf(“Enter the student name: \n”);
 scanf(“%s”,&s.sname);
printf(“Enter the three subjects marks: \n”);
scanf(“%d%d%d”,&s.m1,&s.m2,&s.m3);
 tot = s.m1 + s.m2 +s.m3; avg = tot/3.0;

printf(“Roll Number : %d\n”,s.rno); printf(“Student Name: %s\n”,s.sname);
printf(“Total Marks : %d\n”,tot); printf(“Average : %f\n”,avg);
}
```

Program to read Item Details and Calculate Total Amount of Items

```
#include<stdio.h>
main()
{
struct item
{
int itemno;
char itemname[20]; float rate;
int qty;
};
struct item i; float tot_amt;
 printf(“Enter the Item Number \n”);
scanf(“%d”,&i.itemno); printf(“Enter the Item Name \n”);
scanf(“%s”,&i.itemname);
printf(“Enter the Rate of the Item \n”);
scanf(“%f”,&i.rate);
printf(“Enter the number of %s purchased ”,i.itemname);
scanf(“%d”,&i.qty); tot_amt = i.rate * i.qty;
printf(“Item Number: %d\n”,i.itemno); printf(“Item Name: %s\n”,i.itemname);
printf(“Rate: %f\n”,i.rate); printf(“Number of Items: %d\n”,i.qty); printf(“Total
Amount: %f”,tot_amt);
}
```

ARRAY OF STRUCTURES :

- To store more number of Structures, we can use array of Structures. In array of Structures all elements of the array are stored in adjacent memory location.

Program to illustrate the usage of Array of Structures.

```
#include<stdio.h>
main()
{
struct book
{
char name[20];
 float price;
int pages;
};
struct book b[10];
 int i;
for (i=0;i<10;i++)
{
print("\n Enter Name, Price and Pages");
scanf("%s%f%d", b[i].name,&b[i].price,&b[i].pages);
}
for (i i=0;i<10;i++)
printf(" \n%s%f%d", b[i].name,b[i].price,b[i].pages);
}
```

UNIONS

UNIONS :

Union, similar to Structures, are collection of elements of different data types. However, the members within a union all share the same storage area within the computer‟s memory, whereas each member within a Structure is assigned its own unique Storage area.

Structure enables us to treat a member of different variables stored at different places in memory, a union enables us to treat the same space in memory as a number of different variables. That is, a union offers a way for a section of memory to be treated as a variable of one type on one occasion and as a different variable of a different type on another occasion.

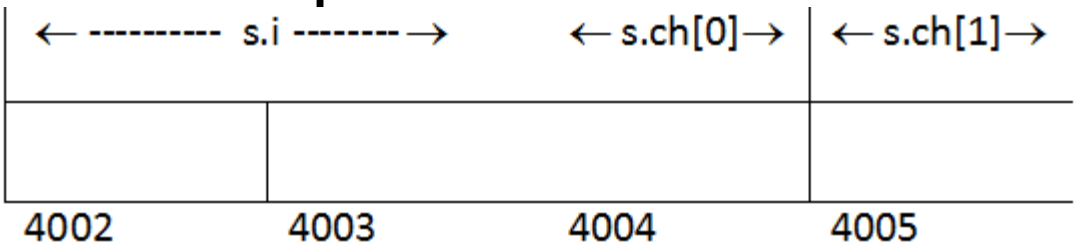
Unions are used to conserve memory. Unions are useful for applications involving multiple members, where values need not be assigned to all of the members at any given time. An attempt to access the wrong type of information will produce meaningless results.

The union elements are accessed exactly the same way in which the structure elements are accessed using dot(.) operator.

The difference between the structure and union in the memory representation is as follows.

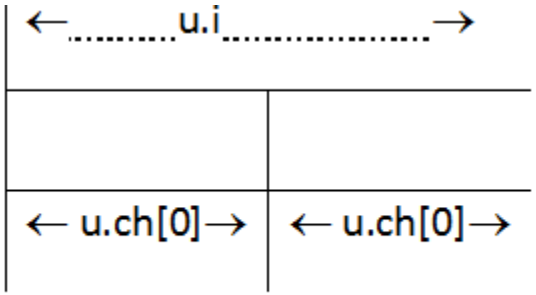
```
struct cx
 {
 int i;
 char ch[2];
 };
 struct cx s1;
```

The Variable s1 will occupy 4 bytes in memory and is represented as



The above datatype, if declared as union then

```
 union ex
{
int i;
char ch[2];
}
union ex u;
```



Program to illustrate the use of unions

```
#include<stdio.h>
main()
{

union example

{

int i;

char ch[2];

};

union exaple u;


u.i = 412;


print(“\n u.i = %d”,u.i);

print(“\n u.ch*0+ = %d”,u.ch*0+); print(“\n u.ch*1+ = %d”,u.ch*1+);


u.ch[0] = 50; /* Assign a new value */


print(“\n\n u.i = %d”,u.i);

print(“\n u.ch*0+ = %d”,u.ch[0]); print(“\n u.ch*1+ = %d”,u.ch*1+);
}
```

Output :

u.i = 512 u.ch[0] = 0 u.ch[1] =2
u.i = 562 u.ch[0] = 50 u.ch[1] =2

A union may be a member of a structure, and a structure may be a member of a union. And also structures and unions may be freely mixed with arrays.

Program to illustrate union with in a Structure

```
#include<stdio.h>
main()
{

union id

{


char color; int size;


};

struct {

char supplier[20]; float cost;
union id desc;


}pant, shirt;

printf("%d\n", sizeof(union id)); shirt.desc.color = 'w';
printf("%c %d\n", shirt.desc.color, shirt.desc.size); shirt.desc.size = 12;
printf("%c %d\n", shirt.desc.color, shirt.desc.size);
 }
```

Binary File Handling

- Binary File Handling is a process in which we create a file and store data in its original format. It means that if we stored an integer value in a binary file, the value will be treated as an integer rather than text.
- Binary files are mainly used for storing records just as we store records in a database. It makes it easier to access or modify the records easily.

Contd..

Mode	Description
rb	Open file in binary mode for reading only.
wb	Open file in binary mode for writing only. It creates the file if it does not exist. If the file exists, then it erases all the contents of the file.
ab	Open file in binary mode for appending data. Data is added to the end of the file. It creates the file if it does not exist.
rb+	Open file in binary mode for both reading and writing.
wb+	Open file in binary mode for both reading and writing. It creates the file if it does not exist. If the file exists, then it erases all the contents of the file.
ab+	Open a file in binary mode for reading and appending data. Data is added to the end of the file. It creates the file if it does not exist.

Reading and Writing a File

- For reading and writing to a text file, we use the functions fprintf() and fscanf().
- They are just the file versions of printf() and scanf(). The only difference is that fprintf() and fscanf() expects a pointer to the structure FILE.

Writing a binary file using fwrite()

```
#include <stdio.h>
#include <stdlib.h>
 int main()
{
   int num;
   FILE *fptr;

   // use appropriate location if you are using MacOS or Linux
   fptr = fopen("C:\\program.txt","w");

   if(fptr == NULL)
   {
      printf("Error!");
      exit(1);
   }

   printf("Enter num: ");
   scanf("%d",&num);

   fprintf(fptr,"%d",num);
   fclose(fptr);

   return 0;
}
```

This program takes a number from the user and stores in the file program.txt.

After you compile and run this program, you can see a text file program.txt created in C drive of your computer. When you open the file, you can see the integer you entered.

Read from a binary file using fread()

```
include <stdio.h>
#include <stdlib.h>

struct threeNum
{
   int n1, n2, n3;
};

int main()
{
   int n;
   struct threeNum num;
   FILE *fptr;

   if ((fptr = fopen("C:\\program.bin","rb")) == NULL){
       printf("Error! opening file");
```

Contd..

```
 // Program exits if the file pointer returns NULL.
      exit(1);
   }

   for(n = 1; n < 5; ++n)
   {
     fread(&num, sizeof(struct threeNum), 1, fptr);
     printf("n1: %d\tn2: %d\tn3: %d\n", num.n1, num.n2, num.n3);
   }
   fclose(fptr);

   return 0;
}
```

In this program, you read the same file program.bin and loop through the records one by one.